Can Community Service Foster Social Agency Among First-Year College STEM Students?

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1 Introduction

Science and technology development has tremendous influence in shaping society, such as reducing environmental pollution, speeding up disease treatment, and connecting people worldwide through digital platforms. However, the benefits of these advances have not been distributed evenly. For example, examinations of medical technology have found that medical AI algorithms exhibit racial and ethnic bias, with Black patients being diagnosed as significantly sicker than White patients at a given risk score (Obermeyer et al., 2019). The mainstream view of STEM education mainly emphasizes two goals: increasing U.S. economic and technological competitiveness and achieving personal financial success (McGee & Bentley, 2017; Riley et al., 2014). A sometimes-overlooked purpose of education is fostering broader social progress and social advancement (Bowen, 1977).

An increasing number of scholars and policymakers have recognized the importance of social justice and socially responsible perspectives in STEM education. Previous studies have highlighted the transformative potential of STEM education in promoting social equity and human good (Bielefeldt & Canney, 2016; Brown et al., 2015; Garibay, 2024; Gutstein, 2012; Martin et al., 2021; McGee & Bentley, 2017; Riley et al., 2014). This shift is reflected in recent policy. The National Science Foundation (NSF, 2025) has encouraged scientists and engineers to conduct research that has broader "potentials to benefit society and contribute to the achievement of specific, desired societal outcomes" (p. 5). However, recent studies show that undergraduate STEM students' social awareness remains flat during college (Schiff et al., 2025), and students committed to social justice tend to leave STEM pathways more frequently when they perceive STEM careers as conflicting with their values and life goals (McGee & Bentley, 2017). Scholars further advocate for expanding STEM education's purpose and success measures to include social justice and socially responsible outcomes, such as civic awareness, social agency, and conducting research for social change (Garibay, 2015, 2018, 2024).

Community service has emerged as a promising strategy that promotes academic and civic development among college STEM students (Bringle & Hatcher, 1996; Narong & Hallinger, 2023). While research has established positive associations between students' development of socially responsible outcomes (such as social agency, civic awareness, and career concerns for social change) and their participation in community service, service learning, and volunteer activities (Einfeld & Collins, 2008; Garibay, 2015; McGee & Bentley, 2017; Museus & Sifuentez, 2021; Sax, 2004), we lack rigorous empirical evidence on whether and to what extent community service activities in universities promote STEM students' social agency. Moreover, the majority of the existing evidence base on community service and STEM social agency is built on frequentist regression and propensity-score methods.

This paper applies a Bayesian framework to examine the association between frequent college community service and senior-year social agency among college STEM students. Using college records from the Higher Education Research Institute's Freshman Survey (TFS, 2021) and College Senior Survey (CSS, 2025), linked to institutional indicators from the Integrated Postsecondary Education Data System (IPEDS), the analytic sample comprises 5,715 STEM undergraduates across 53 institutions. The following research questions guide the analysis:

RQ1: After adjusting for student background, pre-college STEM exposure, baseline social agency, aspirations, and institutional context, what is the posterior association between frequent community service participation during college and senior-year social agency among STEM aspirants?

RQ2: Which student- and institution-level characteristics show the strongest credible associations with senior-year social agency, after conditioning on community service participation?

RQ3: Does the magnitude of the community-service association vary by gender, race/ethnicity, or first-generation status when demographic interactions are estimated under a regularized horseshoe prior?

RQ4: Across eight Bayesian specifications combining linear versus additive functional form, presence versus absence of institutional random intercepts, and default versus regularized horseshoe priors, which offers the best out-of-sample predictive performance under LOO-CV, and how robust is the community-service association across specifications?

2 Methods

2.1 Data and Sample

The data combine longitudinal student records from the HERI's 2021 TFS and 2025 CSS, linked at the student level to institutional indicators from the IPEDS. The TFS, administered at college entry, captures students' background characteristics, pre-college experiences, and psychosocial traits; the CSS, administered toward the end of senior year, captures college experiences, psychosocial outcomes, and post-college plans. The analytic file pairs the TFS 2021 cohort with its corresponding CSS responses in 2025. After restricting to students who reported a STEM major aspiration on the TFS and who completed both surveys, the analytic sample comprises 5,715 students nested in 53 institutions.

2.2 Variables

Outcome. The outcome is social agency, drawn from the HERI CSS. The construct is an IRT-scored composite of items capturing students' belief in their personal capacity to effect social change. Because the IRT scoring produces a small mass point at the upper boundary of the scale, observations within $\varepsilon = 0.05$ of the sample maximum were jittered with independent Gaussian noise $\mathcal{N}(0, 0.10^2)$. This light continuity correction affected approximately 2% of the sample. Figure 1 shows the outcome distribution after smoothing.

Focal predictor. The focal predictors are two binary indicators of students' community service engagement during college: one captures whether the student frequently performed community service work, and the other captures whether the student frequently participated in volunteering activities.

Covariates. At the school level, the models adjust for institutional type (4-year College / University), institutional control (Public / Private), selectivity, and Historically Black College and University (HBCU) status. At the student level, covariates capture (a) student background, including gender, race/ethnicity, first-generation status, parental STEM occupation, high-school GPA, and intended STEM major subfield; (b) pre-college STEM exposure, measured as years of math, physics, biology, and computer science coursework; (c) aspirations and psychosocial traits, comprising highest degree aspiration, academic habits of mind, academic self-concept, and social self-concept; and (d) baseline social agency, measured on the Freshman Survey at college entry to adjust for pre-treatment levels of the outcome construct. All continuous variables enter on a standardized scale.

2.3 Analysis

Three features of the Bayesian approach motivate this analysis. First, posterior distributions provide a full uncertainty summary for the community-service association, including the posterior probability that the effect exceeds zero or any substantive threshold. Second, a regularized horseshoe prior (Piironen & Vehtari, 2017) on the parametric coefficients provides principled handling of covariates of mixed substantive relevance, shrinking negligible effects toward zero while leaving large signals essentially unpenalized, and avoids the ad hoc covariate selection that frequentist analyses often rely on. Third, expected log predictive density estimated via PSIS-LOO cross-validation (Vehtari et al., 2017) supplies a formal comparison criterion across the model specifications introduced below.

Outcome density correction. Because the IRT-scored social agency construct produces a sharp ceiling spike at the upper boundary of the scale, observations within $\varepsilon = 0.05$ of the sample maximum were jittered with independent Gaussian noise prior to model fitting:

$$y_{ij}^{\text{smooth}} = y_{ij} + \eta_{ij}, \quad \eta_{ij} \sim \mathcal{N}\Big(0, 0.10^2\Big) \quad \text{for } y_{ij} \geq y_{\max} - \varepsilon. \tag{1}$$

Model specification. Let i index students and j index institutions ($j = 1, \ldots, 53$). Let y_{ij} denote the smoothed outcome, T_{ij} the binary community-service indicator, $\mathbf{x}_{ij}$ the parametric covariate vector, and $\mathbf{z}_{ij}$ the subset of continuous predictors entering as splines in the additive specifications.

The first set of models (M1 through M4) employs the default weakly informative priors implemented in `rstanarm`: autoscaled Normal priors on the coefficients.

The linear model (M1) is:

$$y_{ij} = \beta_0 + \beta_T T_{ij} + \mathbf{x}_{ij}^\top \boldsymbol{\beta} + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2). \tag{2}$$

The additive model (M2) replaces five continuous covariates with penalized thin-plate splines:

$$y_{ij} = \beta_0 + \beta_T \, T_{ij} + \sum_{k=1}^{5} s_k(z_{ij,k}) + \mathbf{w}_{ij}^\top \boldsymbol{\beta} + \varepsilon_{ij}, \tag{3}$$

where $s_k(z) = \sum_{\ell=1}^{9} b_{k\ell} B_{k\ell}(z)$ and $\mathbf{w}_{ij}$ collects the remaining linear covariates.

The mixed model (M3) follows a two-stage hierarchical structure:

$$\text{Within-institution:} \quad y_{ij} = \beta_{0j} + \beta_T T_{ij} + \mathbf{x}_{ij}^\top \boldsymbol{\beta} + \varepsilon_{ij}, \tag{4}$$

$$\text{Between-institution:} \quad \beta_{0j} = \theta_0 + u_j, \qquad u_j \sim \mathcal{N}(0, \tau^2). \tag{5}$$

The intraclass correlation from an empty model is $\rho = \tau^2/(\tau^2 + \sigma^2) = 0.044$.

The additive mixed model (M4) combines the smooths of M2 with the random intercept of M3:

$$y_{ij} = \beta_{0j} + \beta_T T_{ij} + \sum_{k=1}^{5} s_k(z_{ij,k}) + \mathbf{w}_{ij}^\top \boldsymbol{\beta} + \varepsilon_{ij}, \qquad \beta_{0j} \sim \mathcal{N}(\theta_0, \tau^2). \tag{6}$$

The second set of models (M5 through M8) replicates and extends the structures above but replaces the default Normal prior on each parametric coefficient with the regularized horseshoe. Conditional on its local and global scales, each coefficient is:

$$\beta_p \mid \lambda_p, \tau_g, c \;\sim\; \mathcal{N}\Big(0,\; \tau_g^2\, \tilde{\lambda}_p^2\Big), \qquad \tilde{\lambda}_p^2 = \frac{c^2\, \lambda_p^2}{c^2 + \tau_g^2\, \lambda_p^2}, \tag{7}$$

with local scales $\lambda_p \sim$ Half-Cauchy$(0, 1)$, global scale $\tau_g \sim$ Half-Cauchy$(0, \tau_0)$, and slab scale $c^2 \sim$ Inverse-Gamma$(2, 8)$ (corresponding to slab df = 4 and slab scale = 2.5).

In `stan_gamm4`, the horseshoe prior applies only to the parametric coefficients; the penalized basis priors on the smooths are left unchanged. Models M5, M6, and M7 mirror the linear (M1), additive (M2), and mixed (M3) specifications respectively, with the horseshoe replacing the default coefficient prior. Model M8 further extends the mixed-horseshoe model M7 by adding interactions between frequent community service participation and three demographic dimensions selected to test heterogeneity claims central to the broader research program:

$$y_{ij} = \beta_{0j} + \beta_T T_{ij} + \sum_{g \in \mathcal{G}} \boldsymbol{\beta}_{T:g}^\top (T_{ij} \times G_{ij,g}) + \mathbf{x}_{ij}^\top \boldsymbol{\beta} + \varepsilon_{ij}, \tag{8}$$

with $\mathcal{G} = \{\text{gender, race, firstgen}\}$ and the horseshoe prior of equation (7) applied to all parametric coefficients, including the interaction terms.

Convergence diagnostics. All eight models were estimated with Hamiltonian Monte Carlo via `rstanarm`, using four chains of 2,000 iterations each for a total of 4,000 posterior draws per model. The target acceptance rate δ_{adapt} was set to 0.99 for the additive and horseshoe models to accommodate their heavier-tailed posterior geometry. Convergence was assessed by the Gelman–Rubin potential scale reduction factor $\widehat{R}$ and the bulk effective sample size S_{eff}, with $\widehat{R} \leq 1.01$ and $S_{\text{eff}} \geq 1{,}000$ treated as evidence of adequate mixing.

Posterior predictive checks (PPC). For each fitted model, 100 replicate datasets $\tilde{y}^{(s)}$ were drawn from the posterior predictive distribution and overlaid on the observed outcome density to assess marginal fit:

$$p(\tilde{y} \mid y) = \int_{\Theta} p(\tilde{y} \mid \theta)\, p(\theta \mid y)\, d\theta. \tag{9}$$

Model comparison. Models were compared via LOO cross-validation, which estimates the expected log pointwise predictive density:

$$\widehat{\text{elpd}}_{\text{loo}} = \sum_{i=1}^{n} \log p(y_i \mid y_{-i}). \tag{10}$$

Pairwise differences $\Delta\widehat{\text{elpd}}$ were reported alongside their standard errors se_{diff}, and $|\Delta\widehat{\text{elpd}}| > 2 \cdot \text{se}_{\text{diff}}$ was treated as evidence of a meaningful difference between two specifications.

3 Findings

3.1 Sample Description

Table 1 presents the description of the analytic sample, comprising 5,715 STEM-aspiring undergraduates nested within 53 institutions. Students predominantly attended research universities (62.5%) rather than four-year colleges (37.5%) and private (69.8%) rather than public (30.2%) institutions. Women constituted nearly two-thirds of the sample (64.0%), men approximately one-third (33.6%), and genderqueer or gender-nonconforming students 2.3%. With respect to race/ethnicity, the sample was majority White (54.0%), followed by Asian (17.4%), Multiracial (12.1%), Latino (9.1%), and Black (6.5%) students; Native students were essentially absent ($n = 2$). STEM majors were concentrated in the biological and life sciences (36.4%) and health professions (25.8%), with engineering (19.1%), mathematics and computer science (12.5%), and the physical sciences (6.3%) comprising the remainder.

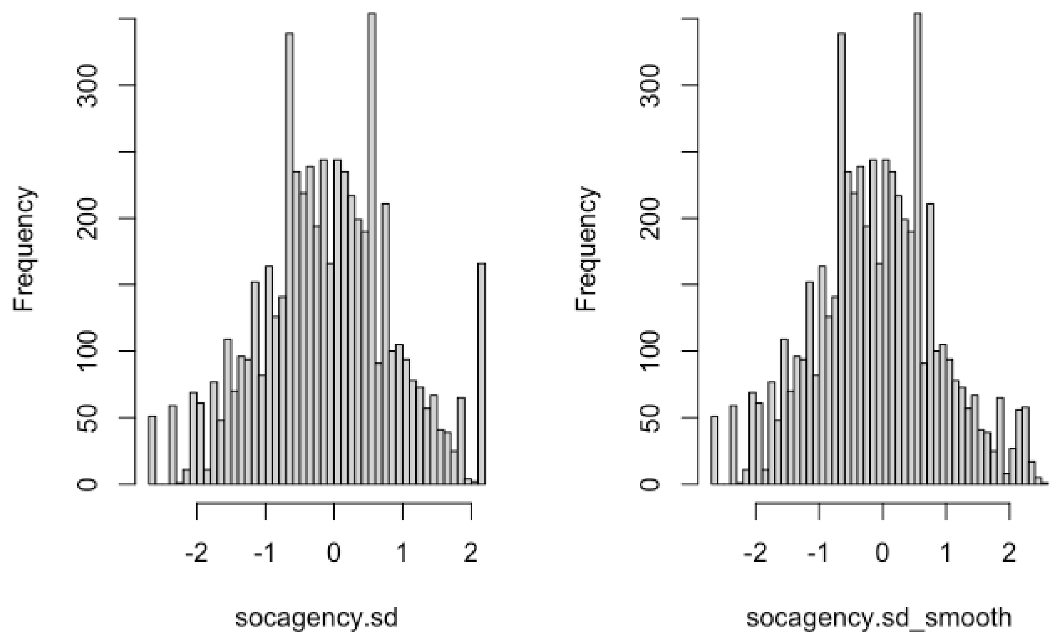

Figure 1: Outcome histogram before and after smoothing.

3.2 Regression Results

I report Model 7 as the main specification in the regression results. Model 7 is a mixed model that combines institution-level random intercepts with a regularized horseshoe prior on the parametric coefficients. I selected this model because it matches the predictive performance of the best-fitting model (M3), while the horseshoe prior additionally provides principled, automatic shrinkage of negligible coefficients, yielding more interpretable estimates across the large covariate set; full details

Table 1: Sample Descriptive Statistics.

Variable	Overall ($N = 5{,}715$)
Social agency	−0.101 (0.990)
Institution type	
4-year College	2,144 (37.5%)
University	3,571 (62.5%)
Institution control	
Public	1,725 (30.2%)
Private	3,990 (69.8%)
Selectivity	0.218 (0.944)
HBCU	
non-HBCU	5,564 (97.4%)
HBCU	151 (2.6%)
Gender	
Men	1,923 (33.6%)
Women	3,659 (64.0%)
Genderqueer/GNC	133 (2.3%)
Race/ethnicity	
White	3,085 (54.0%)
Asian	992 (17.4%)
Black	369 (6.5%)
Latino	518 (9.1%)
Native	2 (0.0%)
Multiracial	690 (12.1%)
Other	59 (1.0%)
STEM major subfield	
Biological & Life Sciences	2,079 (36.4%)
Math & Computer Sciences	713 (12.5%)
Engineering	1,089 (19.1%)
Physical Science	358 (6.3%)
Health Professions	1,476 (25.8%)
First-generation	
No	4,935 (86.4%)
Yes	780 (13.6%)
High-school GPA	0.174 (0.896)
Parent in STEM	
No	3,809 (66.6%)
At least one	1,906 (33.4%)

Note. Continuous variables reported as mean (SD); categorical variables as count (%).

appear in the Model Comparison section below. All coefficients are expressed in outcome SD units, and an association is described as "credible" when its 95% credible interval (CrI) excludes zero (bolded in the results table). The regression results are presented in Table 2.

Service-related activities (RQ1). Frequent community service during college carries a strong positive association with senior-year social agency: posterior mean 0.409 SD, 95% CrI $[0.356, 0.465]$, excluding zero. To put 0.41 SD in perspective, the contrast between students who frequently perform community service and those who do not is slightly larger than the gain associated with a full one-standard-deviation increase in students' incoming (baseline) social agency (0.374 SD). By contrast, the second service-related activity indicator, frequent volunteering, shows no credible association (-0.008, 95% CrI $[-0.039, 0.031]$). After adjusting for student background, pre-college STEM exposure, baseline social agency, aspirations, and institutional context, frequent community-service participation has a credibly positive posterior association with senior-year social agency of about 0.41 SD (95% CrI $[0.36, 0.47]$). Frequent volunteering shows no such association.

Other predictors (RQ2). Conditional on community service, the single strongest predictor was baseline social agency measured at college entry (0.374; 95% CrI $[0.349, 0.398]$), indicating that students' incoming dispositions largely persisted through college. Several additional student characteristics were credibly associated with the outcome. Queer/gender-nonconforming students scored higher than men (0.262; $[0.100, 0.415]$), and Latino (0.157; $[0.052, 0.259]$) and Black (0.106; $[0.006, 0.229]$) students scored higher than White students. Three incoming psychosocial scales were also credible: social self-concept (0.050; $[0.020, 0.078]$) and academic habits of mind (0.038; $[0.010, 0.066]$) were positively associated with senior-year social agency, whereas academic self-concept was negatively associated (-0.034; $[-0.065, -0.009]$). With respect to intended major, engineering (-0.122; $[-0.199, -0.046]$) and mathematics/computer science (-0.093; $[-0.184, -0.003]$) students reported lower social agency than students in the biological and life sciences (the reference group).

The variance components confirmed that social agency is a predominantly individual-level outcome: the residual SD was 0.846 $[0.831, 0.862]$, the institution-intercept variance was small (0.010 $[0.003, 0.021]$), and the model-based intraclass correlation was 0.014 (compared with 0.044 from an unconditional model), indicating that only a small share of outcome variance lay between institutions.

Interaction terms (RQ3). Model 8 extended Model 7 by adding interactions between frequent community service and three demographic dimensions—gender, race/ethnicity, and first-generation status—to evaluate whether the focal association varied across subgroups. In this specification, the community-service association for the reference group (White, male, non-first-generation students) was 0.467 SD $[0.364, 0.573]$, consistent with the Model 7 main effect. However, under the regularized horseshoe prior, the magnitude of the community-service association did not vary credibly by gender, race/ethnicity, or first-generation status; no interaction's 95% credible interval excluded zero.

Table 2: Posterior coefficient estimates from the mixed model with regularized horseshoe prior predicting senior-year STEM students' social agency.

	Mean	**SD**	**2.5%**	**Median**	**97.5%**
(Intercept)	−0.282	**0.078**	−0.439	−0.278	−0.136
Baseline social agency	**0.374**	**0.013**	**0.349**	**0.374**	**0.398**
Frequent community service	**0.409**	**0.028**	**0.356**	**0.409**	**0.465**
Volunteering	−0.008	0.018	−0.039	−0.010	0.031
Institution type: University (ref: college)	0.018	0.029	−0.043	0.019	0.072

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Table 2 continued

	Mean	**SD**	**2.5%**	**Median**	**97.5%**
Institution control: Private (ref: public)	0.087	0.057	−0.016	0.085	0.202
Selectivity	−0.001	0.024	−0.054	0.002	0.037
HBCU	−0.016	0.088	−0.172	−0.024	0.177
Gender: Women (ref: men)	0.058	0.029	−0.002	0.059	0.113
Gender: queer/GNC (ref: men)	**0.262**	**0.081**	**0.100**	**0.262**	**0.415**
Race: Asian (ref: White)	−0.018	0.038	−0.091	−0.019	0.058
Race: Black (ref: White)	**0.106**	**0.060**	**0.006**	**0.103**	**0.229**
Race: Latino (ref: White)	**0.157**	**0.053**	**0.052**	**0.157**	**0.259**
Race: Native (ref: White)	−0.143	0.407	−1.034	−0.090	0.662
Race: Multiracial (ref: White)	0.033	0.032	−0.016	0.027	0.105
Race: Other (ref: White)	0.082	0.094	−0.065	0.065	0.288
Major: Math & Computer Sci (ref: bio)	−0.093	**0.047**	−0.184	−0.093	−0.003
Major: Engineering (ref: bio)	−0.122	**0.039**	−0.199	−0.122	−0.046
Major: Physical Science (ref: bio)	−0.063	0.051	−0.172	−0.060	0.022
Major: Health Professions (ref: bio)	−0.035	0.033	−0.102	−0.034	0.020
First-generation students	−0.036	0.036	−0.112	−0.033	0.021
High-school GPA	0.002	0.011	−0.023	0.003	0.024
At least one parent in STEM field	0.016	0.019	−0.015	0.012	0.062
Years of math	−0.003	0.009	−0.023	−0.003	0.017
Years of physics	0.000	0.009	−0.016	−0.001	0.020
Years of biology	0.019	0.013	−0.004	0.019	0.044
Years of computer science	−0.016	0.012	−0.040	−0.015	0.003
Degree aspiration: Master's (ref: Bachelor's or below)	0.026	0.028	−0.023	0.024	0.086
Degree aspiration: Medical doctorate (ref: Bachelor's or below)	0.033	0.035	−0.026	0.031	0.108
Degree aspiration: Ph.D./Prof. doctorate (ref: Bachelor's or below)	0.034	0.039	−0.025	0.030	0.115
Degree aspiration: Other (ref: Bachelor's or below)	0.010	0.076	−0.128	−0.004	0.190
Academic habits	**0.038**	**0.014**	**0.010**	**0.038**	**0.066**
Academic self-concept	−0.034	**0.015**	−0.065	−0.033	−0.009
Social self-concept	**0.050**	**0.015**	**0.020**	**0.050**	**0.078**
Variance components					
Residual SD (σ)	0.846		0.831	0.846	0.862
Institution intercept variance	0.010		0.003	0.009	0.021
ICC	0.014				

3.3 Model Comparison

Table 3 presents the model comparison results. All eight specifications converged satisfactorily: maximum $\widehat{R} \leq 1.01$; adequate bulk effective sample sizes, with a minimum of approximately 824 in M2; and all Pareto-k values < 0.7, indicating reliable LOO estimates. Models were compared on expected log predictive density (elpd) via PSIS-LOO cross-validation, reported as differences from the best-performing model with accompanying standard errors; a difference was treated as meaningful only when its magnitude exceeded approximately twice its standard error.

The mixed model with default priors (M3) attained the highest elpd and served as the reference. Two models were statistically indistinguishable from it: the mixed model with the horseshoe prior (M7; elpd_diff $= -1.5$, SE $= 3.2$, $|\text{diff}|/\text{SE} = 0.47$) and the horseshoe-plus-interactions model (M8; -3.0, SE $= 4.0$, $|\text{diff}|/\text{SE} = 0.75$). These three specifications form an optimal group that cannot be separated on predictive grounds. A second tier, the additive and linear models (M6, M2, M5, M1), performed approximately 12 to 12.5 elpd units lower, a modest-to-borderline decline. The clearly inferior specification was the additive-mixed model under default priors (M4), which was decisively worse (elpd_diff = −67, SE = 6.7, |diff|/SE = 10) and exhibited severe overfitting: its effective number of parameters inflated to $p_{\text{loo}} = 128$ when penalized splines were combined with random intercepts under default priors.

Moreover, two patterns emerge from this comparison. First, the institution random intercept constituted the single largest source of predictive gain, improving elpd by approximately 12.5 units relative to the linear model (M3 vs. M1). Second, the penalized smooths contributed essentially nothing: M2 performed comparably to M1, and the fitted smooth terms from the additive model were effectively linear across the observed range of each predictor, as presented in Figure 2. There was thus no nonlinearity for the additive terms to capture, and combining smooths with random intercepts merely invited overfitting (M4). Because M7 matched the best model's predictive performance while additionally providing principled variable selection, it was retained as the preferred specification for interpretation despite M3's marginally higher raw elpd.

Finally, Figure 3 displays the posterior distribution of the community-service coefficient across all eight specifications. The posteriors for Models 1 through 7 are nearly indistinguishable, centered at approximately 0.41 SD regardless of functional form, random-effects structure, or prior choice, providing visual confirmation that the focal association is insensitive to modeling decisions.

Table 3: Model comparison and convergence.

Model	**Specification**	**Prior on coefs**	**elpd_diff**	**se_diff**	**|diff|/se**
M3 Mixed	Linear +(1 | UNITID)	Default (Normal)	0	0	—
M7 Mixed + HS	Linear +(1 | UNITID)+ horseshoe	Regularized HS	−1.5	3.2	0.47
M8 Mixed + HS + Int	M7 + comserv × {gender, race, firstgen}	Regularized HS	−3	4	0.75
M6 Additive + HS	Smooths + horseshoe on linear terms	Regularized HS	−12	6.4	1.88
M2 Additive	Smooths, no random effect	Default (Normal)	−12.1	5.6	2.16
M5 Linear + HS	Linear + horseshoe	Regularized HS	−12.5	6.3	1.98
M1 Linear	All covariates linear, no random effect	Default (Normal)	−12.5	5.5	2.27
M4 Additive mixed	Smooths +(1 | UNITID)	Default (Normal)	−67	6.7	10

3.4 Sensitivity to Prior Specification

The eight models were organized into two sets differing only in the prior placed on the parametric coefficients: the first set (M1–M4) employed `rstanarm`'s default weakly informative Normal priors, whereas the second set (M5–M8) substituted the regularized horseshoe. Holding model structure fixed, swapping the prior altered predictive performance negligibly: linear M1 (−12.5) versus horseshoe M5 (−12.5); additive M2 (−12.1) versus horseshoe M6 (−12.0); and mixed M3 (reference) versus horseshoe M7 (−1.5). Each default-versus-horseshoe pair differed by at most 1.5 elpd units, with |diff|/SE well below 1, indicating that the two prior families predicted equally well.

The contribution of the horseshoe prior therefore lies not in predictive accuracy but in interpretability. Across the many covariates of uncertain relevance, the horseshoe shrank negligible coefficients toward zero—including pre-college STEM learning, degree aspirations, high-school GPA, parental STEM

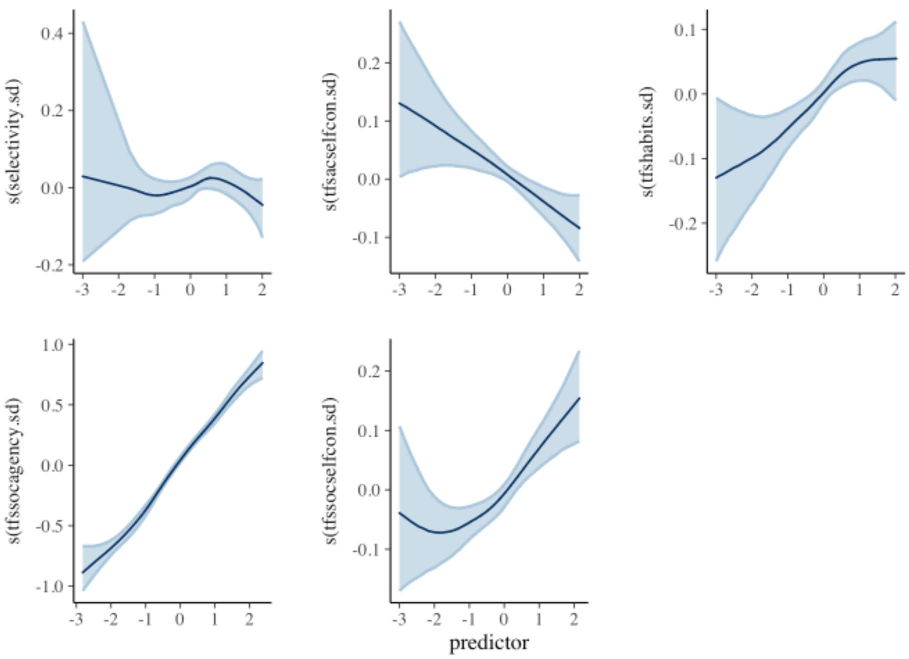

Figure 2: Estimated smooth terms from the additive model.

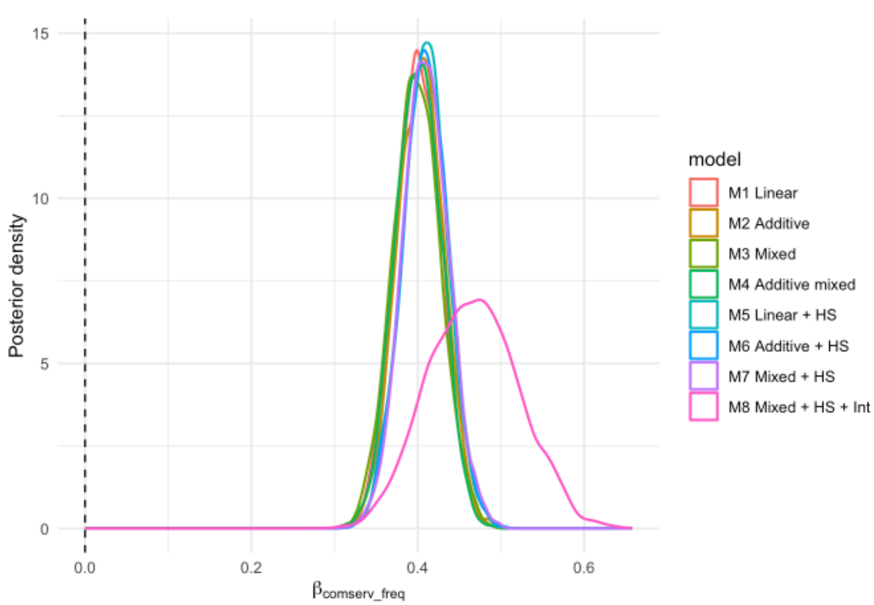

Figure 3: Posterior distributions of the community-service coefficient across all models.

occupation, and most institutional terms, which all collapsed to near zero with intervals straddling zero—while leaving the genuinely large signals, frequent community service (∼0.41 SD) and baseline social agency (∼0.37 SD), essentially unpenalized. The horseshoe thus delivered a clean, automatic separation of signal from noise and obviated the ad hoc covariate selection that frequentist analyses typically require, at no cost to model fit.

4 Discussion

This study examined whether community service participation during college is associated with STEM students' social agency at graduation, and whether that association varies across student subgroups. Using longitudinal HERI data on 5,715 STEM-aspiring undergraduates nested in 53 institutions, I fit eight Bayesian regression specifications varying in functional form, random-effects structure, and prior choice. The headline finding is strong and robust: frequent community service is associated with a 0.41 SD increase in senior-year social agency, an association that stays within 0.40 to 0.47 SD and remains credibly positive across every specification. The magnitude is notable, slightly exceeding the gain linked to a full standard deviation of incoming social agency. In other words, what students do during college appears to matter as much as the disposition they arrive with. The association also appears broadly homogeneous: no interaction with gender, race/ethnicity, or first-generation status reached credibility, which is encouraging from an equity standpoint, since community service appears to relate to social agency gains across demographic groups rather than benefiting some at the expense of others.

Equally informative is what did not predict the outcome. Frequent volunteering, a conceptually adjacent measure, showed no credible association. This divergence suggests that not all prosocial activity operates equivalently: structured community service work may provide the sustained, relational engagement that reshapes students' sense of responsibility for social change, while episodic volunteering may not. This pattern is consistent with the service-learning literature, which finds that the reflective and relational components of service, rather than hours logged, drive civic outcomes. Beyond the focal predictor, minoritized students (genderqueer/gender-nonconforming, Latino, and Black students) reported credibly higher social agency, while engineering and math/computer science majors reported credibly lower social agency than biological science majors, echoing prior evidence that disciplinary cultures differ in the room they make for communal goals. The small intraclass correlation (0.014) indicates that social agency is overwhelmingly an individual-level outcome; institutions explain little variance once student characteristics are accounted for.

The results also raise some statistical issues that shape interpretation. First, the IRT-scored outcome exhibited a ceiling artifact, which I addressed by jittering the clustered maximum values. The posterior predictive check in Figure 4 confirms this restored distributional fit, but the fix treats the symptom rather than the cause: the instrument cannot discriminate among students with the highest social agency, so the model is least informative exactly where the construct is strongest. Second, the null findings are partly prior-dependent in a way the positive findings are not. The regularized horseshoe deliberately shrinks weak signals toward zero, so borderline terms such as the women coefficient and the community-service-by-women interaction (each carrying roughly 93 to 95% posterior probability of being nonzero) might cross conventional thresholds under flatter priors. I view this conservatism as a feature, but it should be made explicit. Third, the model comparison shows that pairing penalized smooths with random intercepts under default priors (M4) produced severe overfitting ($p_{\text{loo}} = 128$), even though each component is harmless alone. Since the fitted smooths were effectively linear, the additive terms had no signal to capture and absorbed noise instead.

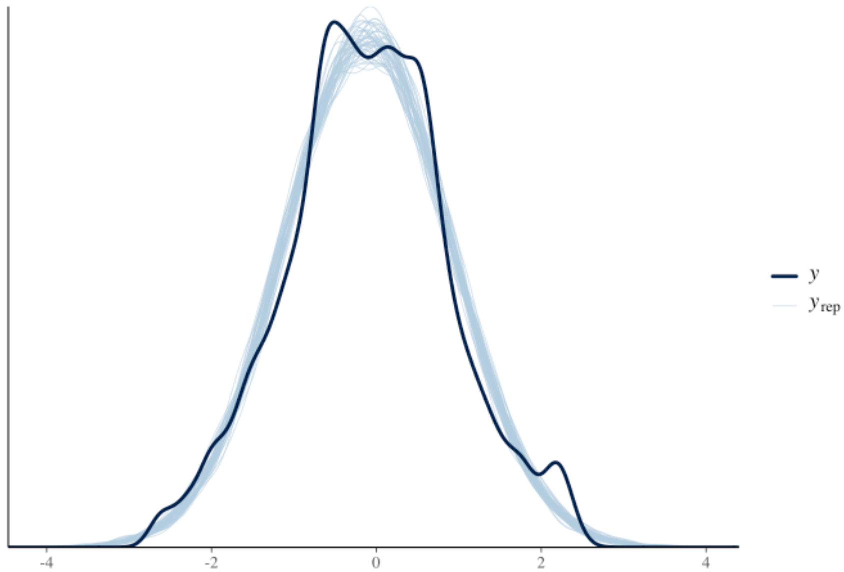

Figure 4: Posterior predictive check for the social agency outcome.

4.1 Limitations

The central limitation is that this is an observational study, and no amount of covariate adjustment converts it into a causal design. Students who frequently perform community service differ from their peers in unmeasured ways, such as the depth of pre-college civic socialization or major-specific service requirements. Adjusting for baseline social agency removes one confounding pathway, but selection on unobservables remains, so the 0.41 SD estimate is an adjusted association, not an effect. Second, the predictor captures self-reported frequency only, with no information on the duration, setting, or curricular embedding of the service. More detailed information on contextual community service could provide a better understanding of how participating in such activities affects students' social agency. Third, the sample overrepresents private institutions, women, and White students, limiting generalizability to community college contexts and underrepresented students. Fourth, with only 53 institutions and minimal between-institution variance, null findings for institutional covariates should be read as not detectable here rather than absent. Finally, both the predictor and outcome come from the same senior-year instrument, raising the possibility of common-method variance.

4.2 Implications for Future Research

There are several directions in which future research can extend this analysis. Future research is encouraged to model the ceiling in the outcome directly, such as by adopting a censored-outcome likelihood that treats the clustered maximum values as right-censored (Tobin, 1958; Wang et al., 2008). This approach would serve as a valuable sensitivity check on the jittering procedure used in this study. In addition, future analyses could allow the community-service association to vary across institutions by incorporating a random slope, examining whether certain campus contexts amplify or constrain the relationship between service participation and social agency. To strengthen causal interpretation, scholars could incorporate propensity score or doubly robust approaches that use baseline covariates to model students' selection into community service (Bang & Robins, 2005; Rosenbaum & Rubin, 1983). Finally, collecting richer measures of the service experience itself is critical for understanding the mechanisms underlying the focal association, including its type, intensity, and whether it was curricular or extracurricular, as prior research suggests that the quality and structure of service, rather than mere participation, drive civic outcomes.

4.3 Conclusions

This study demonstrates that frequent participation in community service is strongly and robustly associated with STEM undergraduates' social agency at the time of graduation, even after accounting for relevant demographic, academic, psychosocial, and institutional factors. The stability of this association across eight model specifications, and its broad homogeneity across gender, racial/ethnic, and first-generation subgroups, suggests that structured service engagement, rather than generic volunteering, is the experience most relevant to cultivating students' commitment to social change. Yet the observational design, the ceiling in the outcome measure, and the unmeasured character of the service itself make clear that these models can identify a robust association but cannot yet establish its causal mechanisms. Addressing these questions requires richer measurement of how students engage in service and stronger designs for modeling who selects into it. These findings highlight the importance of both expanding access to structured community-service opportunities in STEM education and understanding the conditions under which such engagement translates into students' sustained commitment to social agency.

References

Bang, H., & Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics, 61*(4), 962–973. https://doi.org/10.1111/j.1541-0420.2005.00377.x

Bielefeldt, A. R., & Canney, N. E. (2016). Changes in the social responsibility attitudes of engineering students over time. *Science and Engineering Ethics, 22*(5), 1535–1551. https://doi.org/10.1007/s11948-015-9706-5

Bowen, H. R. (1977). *Investment in learning: The individual and social value of American higher education*. The Johns Hopkins University Press.

Bringle, R. G., & Hatcher, J. A. (1996). Implementing service learning in higher education. *The Journal of Higher Education, 67*(2), 221–239. https://doi.org/10.1080/00221546.1996.11780257

Brown, E. R., Thoman, D. B., Smith, J. L., & Diekman, A. B. (2015). Closing the communal gap: The importance of communal affordances in science career motivation. *Journal of Applied Social Psychology, 45*(12), 662–673. https://doi.org/10.1111/jasp.12327

Einfeld, A., & Collins, D. (2008). The relationships between service-learning, social justice, multicultural competence, and civic engagement. *Journal of College Student Development, 49*(2), 95–109. https://doi.org/10.1353/csd.2008.0017

Garibay, J. C. (2015). STEM students' social agency and views on working for social change: Are STEM disciplines developing socially and civically responsible students? *Journal of Research in*

Science Teaching, 52(5), 610–632. https://doi.org/10.1002/tea.21203

Garibay, J. C. (2018). Beyond traditional measures of STEM success: Long-term predictors of social agency and conducting research for social change. *Research in Higher Education, 59*(3), 349–381. https://doi.org/10.1007/s11162-017-9470-2

Garibay, J. C. (2024). Latinx STEM students' values toward conducting research for social change: Examining STEM educational experiences, HSIs, and moderating effects. *Journal of Latinos and Education, 23*(1), 384–402. https://doi.org/10.1080/15348431.2022.2144863

Gutstein, E. (2012). *Reading and writing the world with mathematics: Toward a pedagogy for social justice*. Routledge. https://doi.org/10.4324/9780203112946

Martin, D. A., Conlon, E., & Bowe, B. (2021). A multi-level review of engineering ethics education: Towards a socio-technical orientation of engineering education for ethics. *Science and Engineering Ethics, 27*(5), 60. https://doi.org/10.1007/s11948-021-00333-6

McGee, E., & Bentley, L. (2017). The equity ethic: Black and Latinx college students reengineering their STEM careers toward justice. *American Journal of Education, 124*(1), 1–36. https://doi.org/10.1086/693954

Museus, S. D., & Sifuentez, B. J. (2021). Toward a critical social movements studies. In *Higher education: Handbook of theory and research* (pp. 1–47). Springer, Cham. https://doi.org/10.1007/978-3-030-43030-6_4-1

Narong, D. K., & Hallinger, P. (2023). A keyword co-occurrence analysis of research on service learning: Conceptual foci and emerging research trends. *Education Sciences, 13*(4), Article 4. https://doi.org/10.3390/educsci13040339

NSF. (2025). *Perspectives on broader impacts*. https://www.nsf.gov/funding/learn/broader-impacts

Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science, 366*(6464), 447–453. https://doi.org/10.1126/science.aax2342

Piironen, J., & Vehtari, A. (2017). Sparsity information and regularization in the horseshoe and other shrinkage priors. *Electronic Journal of Statistics, 11*(2), 5018–5051. https://doi.org/10.1214/17-EJS1337SI

Riley, D., Slaton, A. E., & Pawley, A. L. (2014). Social justice and inclusion: Women and minorities in engineering. In *Cambridge handbook of engineering education research* (pp. 335–356). Cambridge University Press. https://doi.org/10.1017/CBO9781139013451.022

Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika, 70*(1), 41–55. https://doi.org/10.1093/biomet/70.1.41

Sax, L. J. (2004). Citizenship development and the American college student. *New Directions for Institutional Research, 2004*(122), 65–80. https://doi.org/10.1002/ir.110

Schiff, D. S., Lee, J., Borenstein, J., & Zegura, E. (2025). Influences and inhibitors in STEM undergraduate social responsibility development. *International Journal of STEM Education, 12*(1), 34. https://doi.org/10.1186/s40594-025-00553-3

Tobin, J. (1958). Estimation of relationships for limited dependent variables. *Econometrica, 26*(1), 24–36. https://doi.org/10.2307/1907382

Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing, 27*(5), 1413–1432. https://doi.org/10.1007/s11222-016-9696-4

Wang, L., Zhang, Z., McArdle, J. J., & Salthouse, T. A. (2008). Investigating ceiling effects in longitudinal data analysis. *Multivariate Behavioral Research, 43*(3), 476–496. https://doi.org/10.1080/00273170802285941

A R Code

The full analysis was conducted in R using **rstanarm**. The code below reproduces every model fit, diagnostic, and figure reported above.

```
## ---- setup ----
library(tidyverse)
library(rstanarm)    # Bayesian regression (course's main package)
library(loo)         # LOO / WAIC model comparison
library(bayesplot)   # posterior predictive checks, diagnostics
library(tidybayes)   # tidy posterior summaries
library(posterior)   # as_draws / summarise_draws for diagnostics
library(knitr)
library(dplyr)

# Reproducibility + use all cores for the 4 MCMC chains
options(mc.cores = parallel::detectCores())
set.seed(216)

# Sampler defaults for every fit below (bump iter if ESS/R-hat need it)
CHAINS <- 4
ITER   <- 2000      # 1000 warmup + 1000 sampling per chain by default

## ---- load-data ----
data <- readRDS("/Users/aishuhan/Desktop/PhD Dissertation/Data/HERICSS2021-25_STEMSRO_clean_v01.rds
    ↪ ")
dim(data)
str(data)

## ---- smooth-ceiling ----
# 1. Inspect the spike
max_val <- max(data$socagency.sd, na.rm = TRUE)
eps     <- 0.05
ceiling_mask <- data$socagency.sd >= (max_val - eps)
n_ceiling    <- sum(ceiling_mask, na.rm = TRUE)
cat("Max value:", round(max_val, 3),
    "| Ceiling threshold (>=", round(max_val - eps, 3), "):",
    n_ceiling, "students (",
    round(100 * n_ceiling / nrow(data), 2), "% of sample )\n")

# 2. Jitter only the ceiling values
set.seed(216)
data$socagency.sd_smooth <- data$socagency.sd
data$socagency.sd_smooth[ceiling_mask] <-
  data$socagency.sd[ceiling_mask] + rnorm(n_ceiling, mean = 0, sd = 0.10)

# 3. Before / after comparison
par(mfrow = c(1, 2))
```

```r
hist(data$socagency.sd,        breaks = 40,
     main = "Original socagency.sd", xlab = "socagency.sd")
hist(data$socagency.sd_smooth, breaks = 40,
     main = "Smoothed (jitter on ceiling)", xlab = "socagency.sd_smooth")
par(mfrow = c(1, 1))

# Summary stats: confirm minimal global distortion
summary(data$socagency.sd)
summary(data$socagency.sd_smooth)

## ---- chunk ----
library(table1)
descr <- table1(~ socagency.sd_smooth +
                  # school IVs
                  insttype + instcont + selectivity.sd + hbcu +
                  # student background (gender now 3 levels: Men/Women/GNC)
                  gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b, data = data)
descr

## ---- fit-m1-linear ----
hist(data$socagency.sd_smooth, breaks = 40)

m1_linear <- stan_glm(
  socagency.sd_smooth ~
    # baseline social agency measures
    tfssocagency.sd +
    # community service activity
    comserv_freq + tfsvolunteer +
    # school IVs
    insttype + instcont + selectivity.sd + hbcu +
    # student background (gender now 3 levels: Men/Women/GNC)
    gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b +
    # pre-college STEM
    ymath.sd + yphys.sd + ybio.sd + ycs.sd +
    # aspirations, values, abilities
    degreeasp.sub +
    tfshabits.sd + tfsacselfcon.sd + tfssocselfcon.sd,
  data   = data,
  family = gaussian(),
  chains = 4, iter = 2000, seed = 216, refresh = 0
)

# Coefficient table
summary(m1_linear, probs = c(0.025, 0.5, 0.975), digits = 3)
prior_summary(m1_linear)

## ---- fit-m2-additive ----
# M2 -- Additive: selected continuous predictors enter as penalized splines
# s(); the rest stay linear. stan_gamm4() fits GAMs in rstanarm (it wraps mgcv's smooth construction).
m2_additive <- stan_gamm4(
  socagency.sd_smooth ~
    # baseline social agency measure
    s(tfssocagency.sd) +
    # community service activity
    comserv_freq + tfsvolunteer +
    # school IVs
    insttype + instcont + s(selectivity.sd) + hbcu +
```

```r
    # student background
    gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b +
    # pre-college STEM
    ymath.sd + yphys.sd + ybio.sd + ycs.sd +
    # aspirations, values, abilities
    degreeasp.sub + s(tfshabits.sd) + s(tfsacselfcon.sd) + s(tfssocselfcon.sd),
  data = data, family = gaussian(),
  chains = CHAINS, iter = ITER, seed = 216, refresh = 0,
  adapt_delta = 0.99           # smooths sometimes need a higher target
)

# Coefficient table
summary(m2_additive, probs = c(0.025, 0.5, 0.975), digits = 3)
prior_summary(m2_additive)

## ---- fit-m3-mixed ----
# Calculate ICC first
library(lme4)

m0 <- lmer(socagency.sd_smooth ~ 1 + (1 | UNITID), data = data)
vc     <- as.data.frame(VarCorr(m0))
tau2   <- vc$vcov[vc$grp == "UNITID"]     # between-institution variance
sigma2 <- vc$vcov[vc$grp == "Residual"]   # within-institution (residual) variance
icc    <- tau2 / (tau2 + sigma2)
icc   # ICC = 0.044

# M3 -- Mixed: same linear specification as M1 PLUS a random intercept by
# institution. stan_glmer() is rstanarm's lme4-style multilevel function;
# the group-variance prior is decov() by default.
m3_mixed <- stan_glmer(
  socagency.sd_smooth ~
    # baseline social agency measure
    tfssocagency.sd +
    # community service activity
    comserv_freq + tfsvolunteer +
    # school IVs
    insttype + instcont + selectivity.sd + hbcu +
    # student background
    gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b +
    # pre-college STEM
    ymath.sd + yphys.sd + ybio.sd + ycs.sd +
    # aspirations, values, abilities
    degreeasp.sub + tfshabits.sd + tfsacselfcon.sd + tfssocselfcon.sd +
    # institution random intercept
    (1 | UNITID),
  data = data, family = gaussian(),
  chains = CHAINS, iter = ITER, seed = 216,
  adapt_delta = 0.95
)

# Coefficient table
summary(m3_mixed, probs = c(0.025, 0.5, 0.975), digits = 3)
prior_summary(m3_mixed)

## ---- fit-m4-addmixed ----
# M4 -- Additive mixed: the M2 smooths and a random intercept
m4_addmixed <- stan_gamm4(
  socagency.sd_smooth ~
    # baseline social agency measure
```

```r
    s(tfssocagency.sd) +
    # community service activity (comserv_freq = treatment)
    comserv_freq + tfsvolunteer +
    # school IVs
    insttype + instcont + s(selectivity.sd) + hbcu +
    # student background
    gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b +
    # pre-college STEM
    ymath.sd + yphys.sd + ybio.sd + ycs.sd +
    # aspirations, values, abilities
    degreeasp.sub + s(tfshabits.sd) + s(tfsacselfcon.sd) + s(tfssocselfcon.sd),
  random = ~(1 | UNITID),
  data = data, family = gaussian(),
  chains = CHAINS, iter = ITER, seed = 216,
  adapt_delta = 0.99
)

# Coefficient table
summary(m4_addmixed, probs = c(0.025, 0.5, 0.975), digits = 3)
prior_summary(m4_addmixed)

## ---- hs-spec ----
# Global scale: tau0 = (p0 / (p - p0)) * sigma_y / sqrt(n)
# Use p = 30, p0 = 5; sigma_y on standardized outcome ~ 1
p_total   <- 30
p_nonzero <- 5
n_obs     <- nrow(data)
tau0      <- (p_nonzero / (p_total - p_nonzero)) / sqrt(n_obs)
tau0   # ~ 0.0026; round up to 0.01 for a slightly less aggressive global shrinkage

hs_prior <- hs(df = 1, global_df = 1, global_scale = 0.01,
               slab_df = 4, slab_scale = 2.5)

## ---- fit-m5-linear-hs ----
m5_linear_hs <- stan_glm(
  socagency.sd_smooth ~
    # baseline social agency
    tfssocagency.sd +
    # community service activity
    comserv_freq + tfsvolunteer +
    # school IVs
    insttype + instcont + selectivity.sd + hbcu +
    # student background
    gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b +
    # pre-college STEM
    ymath.sd + yphys.sd + ybio.sd + ycs.sd +
    # aspirations, values, abilities
    degreeasp.sub + tfshabits.sd + tfsacselfcon.sd + tfssocselfcon.sd,
  data   = data,
  family = gaussian(),
  prior  = hs_prior,
  chains = CHAINS, iter = ITER, seed = 216, refresh = 0,
  adapt_delta = 0.99
)

summary(m5_linear_hs, probs = c(0.025, 0.5, 0.975), digits = 3)
prior_summary(m5_linear_hs)

## ---- fit-m6-additive-hs ----
```

```
m6_additive_hs <- stan_gamm4(
  socagency.sd_smooth ~
    s(tfssocagency.sd) +
    comserv_freq + tfsvolunteer +
    insttype + instcont + s(selectivity.sd) + hbcu +
    gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b +
    ymath.sd + yphys.sd + ybio.sd + ycs.sd +
    degreeasp.sub + s(tfshabits.sd) + s(tfsacselfcon.sd) + s(tfssocselfcon.sd),
  data = data, family = gaussian(),
  prior  = hs_prior,          # applies to parametric coefs only; smooths unshrunk
  chains = CHAINS, iter = ITER, seed = 216, refresh = 0,
  adapt_delta = 0.99
)

summary(m6_additive_hs, probs = c(0.025, 0.5, 0.975), digits = 3)
prior_summary(m6_additive_hs)

## ---- fit-m7-mixed-hs ----
m7_mixed_hs <- stan_glmer(
  socagency.sd_smooth ~
    tfssocagency.sd +
    comserv_freq + tfsvolunteer +
    insttype + instcont + selectivity.sd + hbcu +
    gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b +
    ymath.sd + yphys.sd + ybio.sd + ycs.sd +
    degreeasp.sub + tfshabits.sd + tfsacselfcon.sd + tfssocselfcon.sd +
    (1 | UNITID),
  data = data, family = gaussian(),
  prior  = hs_prior,          # fixed-effect coefs; RI keeps decov()
  QR     = TRUE,              # orthogonalize design matrix (faster sampling)
  chains = CHAINS, iter = ITER, seed = 216, refresh = 0,
  adapt_delta = 0.99
)

summary(m7_mixed_hs, probs = c(0.025, 0.5, 0.975), digits = 3)
prior_summary(m7_mixed_hs)

## ---- fit-m8-mixed-hs-int ----
m8_mixed_hs_int <- stan_glmer(
  socagency.sd_smooth ~
    tfssocagency.sd +
    comserv_freq + tfsvolunteer +
    insttype + instcont + selectivity.sd + hbcu +
    gender + race + tfsmajor.stemsub + firstgen.b + hsgpa.sd + parentstem.b +
    ymath.sd + yphys.sd + ybio.sd + ycs.sd +
    degreeasp.sub + tfshabits.sd + tfsacselfcon.sd + tfssocselfcon.sd +
    # demographic interactions with treatment
    comserv_freq:gender + comserv_freq:race + comserv_freq:firstgen.b +
    (1 | UNITID),
  data = data, family = gaussian(),
  prior  = hs_prior,
  QR     = TRUE,              # orthogonalize design matrix (faster sampling)
  chains = CHAINS, iter = ITER, seed = 216, refresh = 0,
  adapt_delta = 0.99
)

summary(m8_mixed_hs_int, probs = c(0.025, 0.5, 0.975), digits = 3)

# Pull out only the interaction coefficients for inspection
```

```r
int_pars <- grep("^comserv_freq.*:", rownames(summary(m8_mixed_hs_int)), value = TRUE)
summary(m8_mixed_hs_int, pars = int_pars,
        probs = c(0.025, 0.5, 0.975), digits = 3)

## ---- diagnostics ----
fits <- list(
  `M1 Linear`           = m1_linear,
  `M2 Additive`         = m2_additive,
  `M3 Mixed`            = m3_mixed,
  `M4 Additive mixed`   = m4_addmixed,
  `M5 Linear + HS`      = m5_linear_hs,
  `M6 Additive + HS`    = m6_additive_hs,
  `M7 Mixed + HS`       = m7_mixed_hs,
  `M8 Mixed + HS + Int` = m8_mixed_hs_int
)

# R-hat and bulk-ESS check for every model (want R-hat ~ 1.00, ESS in 1000s)
# as.matrix() on a stanreg object returns the posterior draws matrix.
purrr::imap_dfr(fits, function(m, nm) {
  s <- posterior::summarise_draws(as.matrix(m), "rhat", "ess_bulk")
  tibble(model = nm,
         max_rhat = max(s$rhat, na.rm = TRUE),
         min_ess  = min(s$ess_bulk, na.rm = TRUE))
}) %>% kable(digits = 3, caption = "Convergence summary across models")

## ---- ppcheck ----
# Posterior predictive checks -- does each model reproduce the outcome dist
pp_check(m1_linear,      ndraws = 100) + ggtitle("M1 Linear")
pp_check(m2_additive,    ndraws = 100) + ggtitle("M2 Additive")
pp_check(m3_mixed,       ndraws = 100) + ggtitle("M3 Mixed")
pp_check(m4_addmixed,    ndraws = 100) + ggtitle("M4 Additive mixed")
pp_check(m5_linear_hs,   ndraws = 100) + ggtitle("M5 Linear + HS")
pp_check(m6_additive_hs, ndraws = 100) + ggtitle("M6 Additive + HS")
pp_check(m7_mixed_hs,    ndraws = 100) + ggtitle("M7 Mixed + HS")
pp_check(m8_mixed_hs_int,ndraws = 100) + ggtitle("M8 Mixed + HS + Int")

## ---- smooth-plots ----
# Visualize the estimated
# plot_nonlinear() is rstanarm's helper for stan_gamm4 smooth term
plot_nonlinear(m2_additive)

## ---- loo-compare ----
loo_m1 <- loo(m1_linear)
loo_m2 <- loo(m2_additive)
loo_m3 <- loo(m3_mixed)
loo_m4 <- loo(m4_addmixed)        # overfitting under default priors
loo_m5 <- loo(m5_linear_hs)
loo_m6 <- loo(m6_additive_hs)
loo_m7 <- loo(m7_mixed_hs)
loo_m8 <- loo(m8_mixed_hs_int)

# loo_compare orders models best-first
loo_compare(loo_m1, loo_m2, loo_m3, loo_m4,
            loo_m5, loo_m6, loo_m7, loo_m8)

print(loo_m4)
print(loo_m8)

## ---- tx-effect-summary ----
```

```r
# Pull posterior draws for the main treatment coefficient from every fit
tx_par <- "comserv_freqTreated"

tx_draws <- purrr::imap_dfr(fits, function(m, nm) {
  d <- as.data.frame(m, pars = tx_par)
  tibble(model = nm, beta = d[[tx_par]])
})

# Posterior summary table
tx_summary <- tx_draws %>%
  group_by(model) %>%
  summarise(
    mean   = mean(beta),
    sd     = sd(beta),
    q025   = quantile(beta, 0.025),
    q975   = quantile(beta, 0.975),
    p_gt_0 = mean(beta > 0)
  )
tx_summary %>% kable(digits = 3,
  caption = "Posterior of community-service association across models")

# Posterior density overlay
ggplot(tx_draws, aes(x = beta, colour = model)) +
  geom_density(linewidth = 0.7) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(x = expression(beta[comserv_freq]),
       y = "Posterior density",
       title = "Posterior of community-service association across models") +
  theme_minimal()

## ---- m8-interactions ----
# Extract interaction-coefficient posteriors and plot 95% CrIs
all_pars <- rownames(summary(m8_mixed_hs_int))
int_pars <- grep("^comserv_freq.*:|^comserv_freqTreated:", all_pars, value = TRUE)
int_pars

int_draws <- as.data.frame(m8_mixed_hs_int, pars = int_pars) %>%
  tidyr::pivot_longer(everything(), names_to = "term", values_to = "beta")

int_draws %>%
  group_by(term) %>%
  summarise(
    mean = mean(beta), sd = sd(beta),
    q025 = quantile(beta, 0.025), q975 = quantile(beta, 0.975),
    p_gt_0 = mean(beta > 0)
  ) %>%
  arrange(desc(abs(mean))) %>%
  kable(digits = 3,
        caption = "M8: posterior of comserv x demographic interaction terms")

ggplot(int_draws, aes(x = beta, y = term)) +
  tidybayes::stat_halfeye(.width = c(0.5, 0.95)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(x = "Interaction coefficient (on standardized outcome)",
       y = NULL,
       title = "M8: heterogeneity of community-service association") +
  theme_minimal()
```